

IN104: Monte Carlo Simulation of the Ising Model

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March 2026

1 Introduction

Monte Carlo simulations are a powerful class of numerical methods used to study complex systems across many fields, including physics, biology, chemistry, quantitative finance, and applied mathematics. The central idea is to rely on randomness to explore a probability distribution and estimate quantities of interest (observables), together with their statistical uncertainties.

In this project, we apply this methodology to a simple yet fundamental model from statistical physics: the Ising model. Despite its simplicity, this model captures the essential mechanisms behind ferromagnetism and provides an ideal framework for learning modern computational techniques.

1.1 Project Overview

Some materials exhibit a ferromagnetic phase, meaning they develop a spontaneous and persistent magnetization due to the collective alignment of microscopic magnetic moments. However, this ordered behavior only exists below a certain temperature, called the critical temperature T_c . Above this temperature, thermal fluctuations destroy the alignment, and the material becomes disordered (paramagnetic phase).

To model this behavior, we introduce a simplified description:

- Each atom is represented by a *spin*
- Each spin can take only two values:

$$S_i = +1 \quad (\text{“up”}) \quad \text{or} \quad S_i = -1 \quad (\text{“down”}) \quad (1)$$

1.1.1 The Ising Model

The energy of a configuration of spins $\{S_i\}$ is defined as:

$$E(\{S_i\}) = -J \sum_{\langle m,n \rangle} S_m S_n \quad (2)$$

where J is the interaction strength and $\langle m,n \rangle$ denotes pairs of nearest neighbors.

When $J > 0$, the energy is minimized when neighboring spins are aligned. At zero temperature ($T = 0$), the system reaches its ground state, in which all spins are aligned (either all $+1$ or all -1). These two configurations are equivalent and correspond to the ferromagnetic phase.

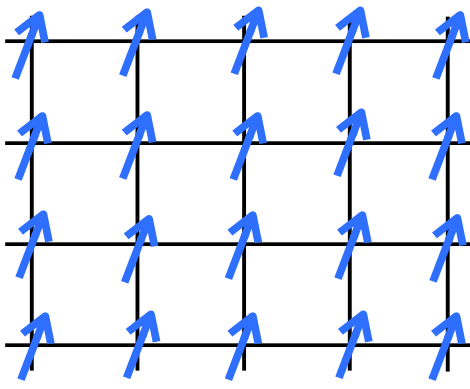
1.1.2 Role of Temperature

At finite temperature, thermal fluctuations introduce disorder, and the system explores multiple configurations. The probability of observing a configuration m , denoted $\{S_i^m\}$, is given by the Boltzmann distribution:

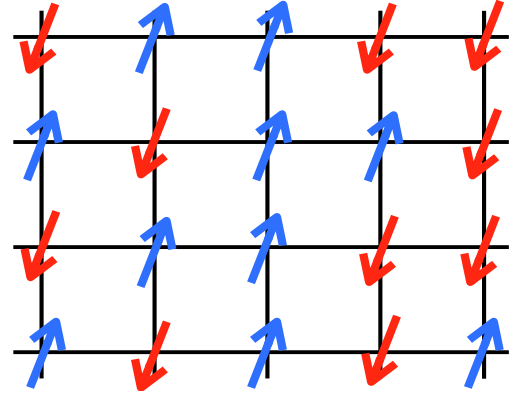
$$P(\{S_i^m\}) = \frac{1}{Z} e^{-E(\{S_i^m\})/T} \quad (3)$$

where Z is the partition function, a normalization constant ensuring that:

$$\sum_m P(\{S_i^m\}) = 1 \quad (4)$$



(a) $T = 0$: ferromagnetic phase, $\langle m \rangle = 1$



(b) High T : paramagnetic phase, $\langle m \rangle \approx 0$

Figure 1: Comparison between ordered (ferromagnetic) and disordered (paramagnetic) phases.

1.2 Methods and Objectives

The goal of this project is to use Monte Carlo methods to sample configurations of the Ising model according to the Boltzmann distribution. To achieve this, we will implement the Metropolis algorithm.

1.2.1 Metropolis Algorithm

The Metropolis algorithm provides an efficient way to sample the Boltzmann distribution. A brute-force approach would require evaluating all possible configurations, which is computationally infeasible: for example, a 10×10 lattice contains 100 spins, corresponding to 2^{100} possible configurations.

The algorithm proceeds as follows:

- (1) Start from an initial configuration $\{S_i^{(0)}\}$
- (2) Compute its energy E_0
- (3) Select a spin at random and propose to flip it
- (4) Compute the new energy E_n
- (5) Accept the move:
 - always if $E_n < E_0$
 - otherwise with probability $e^{-(E_n - E_0)/T}$
- (6) Update the configuration accordingly
- (7) Measure observables and repeat the process (go back to step (3))

Importantly, there is no need to compute the partition function Z , since it cancels when taking ratios of probabilities. Also, It is of paramount importance that observables are computed even if the move have been rejected.

This procedure generates a Markov chain: each new configuration depends only on the current one. Under general conditions (ergodicity and detailed balance), the sampled configurations converge to the desired Boltzmann distribution.

1.3 Computing Observables

From the generated samples, we will compute macroscopic observables such as the magnetization and study their dependence on temperature. In particular, we will identify the phase transition and estimate the critical temperature T_c , which can be compared to the exact analytical result for the two-dimensional Ising model:

$$T_c \approx 2.269 J \quad (5)$$

A second objective of the project is to assess the reliability of Monte Carlo simulations. We will implement post-processing tools to compute:

- mean values and statistical uncertainties
- autocorrelation functions
- binning analysis

These techniques are essential when dealing with correlated data, and allow for a proper estimation of error bars in Monte Carlo simulations.

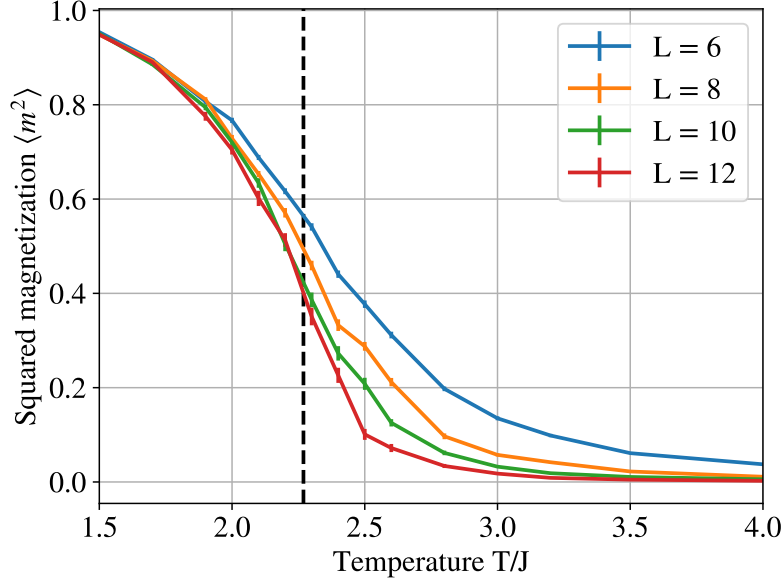


Figure 2: Squared magnetization vs. temperature

1.4 Postprocessing Tools

1.4.1 Observables

Each time we have a configuration, we can calculate the values of different observables. For example we compute the total energy associated to that specific configuration, which is also used to accept or reject the next move. A key quantity that we want to calculate is the average magnetization:

$$m = \frac{1}{N} \sum_m^N S_m \quad (6)$$

where $N = L_x \times L_y = L^2$ is the system size. It is important to store the values of observables and save them to file, such that when the simulation is finished we can compute the mean value and the standard deviation.

It turns out that the average magnetization is not a good quantity to determine if we are in the paramagnetic or ferromagnetic phase: when averaging over all the samples, it may happen that we go from a ferromagnetic phase with negative magnetization to one with positive magnetization. To overcome this problem, we will also store the squared magnetization:

$$m^2 = \left(\frac{1}{N} \sum_m^N S_m \right)^2 \quad (7)$$

Note: the square must be taken for each sample and then we will average these values over all the samples. In Fig. 2, we can see how the squared magnetization changes when tuning temperature, leading to a phase transition from a ferromagnetic to paramagnetic phase.

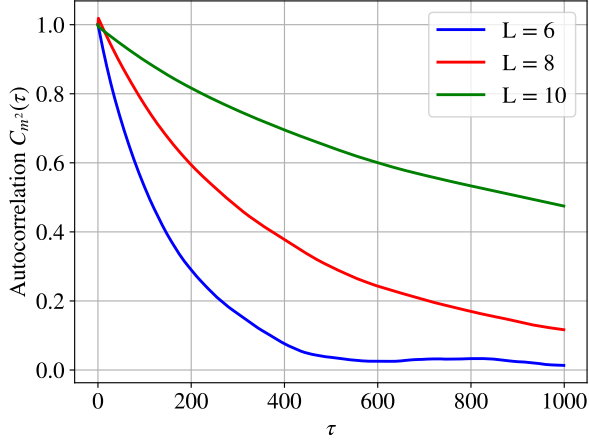
1.4.2 Statistical Errors

In simulations like Monte Carlo, each result must come with an uncertainty. It turns out that taking the standard deviation of all the values of a given observables O usually underestimate the error. The reason is that each configuration is strongly correlated to the previous ones. To quantify this, we compute the autocorrelation function:

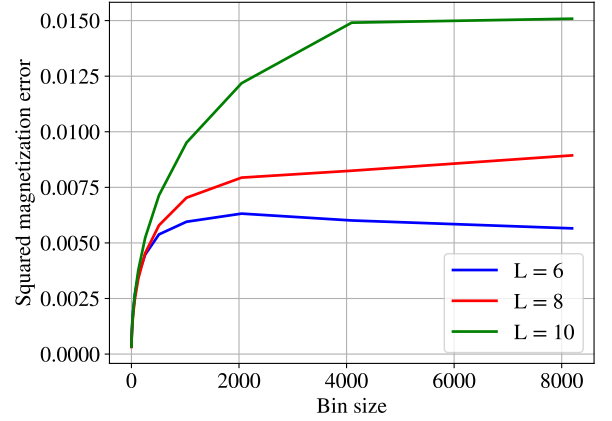
$$C_O(\tau) = \frac{\langle O(t)O(t+\tau) \rangle - \langle O(t) \rangle \langle O(t) \rangle}{\sigma_O^2} \quad (8)$$

In Fig. 3a, a typical autocorrelation function in the Metropolis algorithm is shown.

To overcome this problem, we perform a binning analysis: we group data into bins of different sizes and calculate the mean value of each bin. We then calculate the standard deviation over all bins and plot it as a function of the bin sizes. We see that when the bin size matches with the autocorrelation time, the error saturates to the upper bound, which is the estimation of error. (see fig.3b).



(a) Autocorrelation at $T = 2.3J$, close to the phase transition.



(b) Binning analysis at $T = 2.3J$, close to the phase transition.

Figure 3: Post-processing of Monte Carlo $\langle m^2 \rangle$ observable.

2 Discussion and Presentation of Results

Once we have implemented the Metropolis algorithm, we can run our simulations and extract some results. For example, we can tune the temperature ($T \in (1.5J, \dots, 3.5J)$) for various system sizes ($L \in (6, \dots, 12)$). We will reject the first N_{eq} sampled configurations. This concept is called *thermalization*: we make sure that we don't compute observables before the Metropolis algorithm has converged to the equilibrium distribution. For our system sizes, we will not need large N_{eq} .

The following points are intended as guidelines to help interpret the numerical results, understand the phase transition and the limitations of the algorithm.

- Plot the squared magnetization as a function of temperature; what happens when increasing the system size?
- Plot the total energy as a function of temperature.
- Compute the Binder cumulant

$$b = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2} \quad (9)$$

and plot it as a function of temperature for various system sizes. This quantity is scale invariant (size-independent) at the critical point: can we use this plot to extrapolate the critical temperature?

- Compute the autocorrelation of the squared magnetization at the critical temperature. What happens when increasing the system size? What does it mean?
- Plot the specific heat per spin

$$C_v = \frac{1}{N} \frac{\partial \langle E \rangle}{\partial T} = \frac{\langle E^2 \rangle - \langle E \rangle^2}{NT^2} \quad (10)$$

as a function of temperature for various system sizes.

- Plot the magnetic susceptibility per spin

$$\chi = \frac{1}{N} \frac{\partial \langle M \rangle}{\partial B} = \frac{\langle M^2 \rangle - \langle M \rangle^2}{NT^2} \quad (11)$$

as a function of temperature for various system sizes.

3 EXTRA BONUS: Swendsen-Wang Algorithm

We have seen that Metropolis algorithm is an efficient way to sample the Boltzmann probability distribution and, therefore, calculate observables like magnetization at a given temperature. However, especially when $T \approx T_c$, the autocorrelation function decays very slowly, and it is very difficult to generate configurations that are really independent from the previous ones. This fact makes it difficult to explore all the possible configurations and, therefore, to properly estimate observables. At low T , flipping a single spin costs high energy, and therefore most of the moves are rejected; the Metropolis algorithm finds one of the two macroscopic values of magnetization and gets stuck without exploring the other one. One possible solution is to use cluster algorithms, where multiple spins are flipped at the same time. The Swendsen-Wang algorithm is one possibility:

- (1) Start from an initial configuration $\{S_i^{(0)}\}$
- (2) Compute its energy E_0
- (3) connect neighboring spins i and j based on the following rule:
 - never if $S_i \neq S_j$;
 - otherwise with probability $1 - e^{-2J/T}$

this way, create clusters of connected spins;

- (6) flip each cluster with probability 0.5;
- (7) Measure observables and repeat the process (go back to step (3))

It is possible to show that the sampled configurations follow the Boltzmann probability distribution.

- Implement the Swendsen-Wang algorithm. Hint: an efficient way to identify and merge connected clusters is to use a union-find (disjoint-set) data structure.
See for example https://cp-algorithms.com/data_structures/disjoint_set_union.html.
- Run the algorithm near the critical temperature. Compare the autocorrelation function of the squared magnetization with the one of the Metropolis algorithm. What do you observe?